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SEAT No. :

PE-324

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[6580]-695

B.E. (AI&DS)(Insem)

QUANTUM ARTIFICIAL INTELLIGENCE

(2019 Pattern) (Semester - VII) (417523 A) (Elective-III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q. No. 1 or Q. No. 2, and Q. No. 3 or Q. No. 4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Describe the differences in measurement in bases and computational basis. [5]
- b) Explain Single qubit and multiple qubit gates with diagram. [5]
- c) Demonstrate quantum teleportation with suitable example. [5]

OR

- Q2)** a) Describe Qubit copying circuit. [6]
- b) Draw quantum circuits diagram that represents the operations. [6]
- c) Describe how tensor product combines qubit states? [3]
- Q3)** a) Describe Time-Evolution of a Closed System. [5]
- b) Describe role of quantum entanglement in quantum architecture. [5]
- c) Explain Mixed States and General Quantum Operations. [5]

OR

- Q4)** a) Describe hidden sub group problem for quantum Fourier transform. [5]
- b) Explain universal sets of quantum gates. [5]
- c) Compare order-finding and factoring for Fourier transform. [5]

